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Course: BCA – Cyber Security Lab

Title: Vulnerability Assessment and Penetration Testing (VAPT) Report

Environment: Kali Linux Virtual Lab

Submission Date: 27 February 2026

1. Introduction

This report presents the vulnerability assessment and security analysis performed in a controlled virtual lab environment. The objective of this task was to understand security assessment using open-source tools and identify potential weaknesses in a system without using paid software.

The assessment was performed using Kali Linux and other open-source tools.

2. Theoretical Knowledge

2.1 Understanding Security Assessment

Security Assessment is the process of identifying security weaknesses in a system using structured methods and security frameworks like **National Institute of Standards and Technology (NIST)**.

Types of Security Testing:

- Vulnerability Assessment – Identifies known weaknesses using scanning tools.
- Penetration Testing – Simulates real attacks to test exploitation possibilities.
- Compliance Testing – Ensures systems follow security standards and policies.

Tools Used:

- OpenVAS
- Nmap
- Metasploit Framework



2.2 VAPT Methodology

The following structured phases were followed:

1. Planning – Defined scope and target system.
2. Discovery – Identified open ports and services using Nmap.
3. Vulnerability Identification – Used OpenVAS and Nikto.
4. Reporting – Documented findings and remediation.

Reference Framework:

- OWASP Web Security Testing Framework

2.3 Security Standards & Compliance

Security standards help organizations maintain secure systems.

Important Standards:

- General Data Protection Regulation
- Health Insurance Portability and Accountability Act
- ISO/IEC 27001

OWASP Top 10 was used to understand common web vulnerabilities.

2.4 Risk Assessment Basics

Risk was calculated using:

- CVSS scoring system
- 3x3 Risk Matrix (Likelihood vs Impact)
- Reference: National Vulnerability Database

2.5 Common Vulnerabilities

Network Vulnerabilities:

- Open ports
- Service exposure
- Misconfigurations

Web Vulnerabilities:

- Missing security headers
- Information disclosure



3. Practical Implementation

3.1 Lab Setup

Environment Used:

- VirtualBox
- Kali Linux

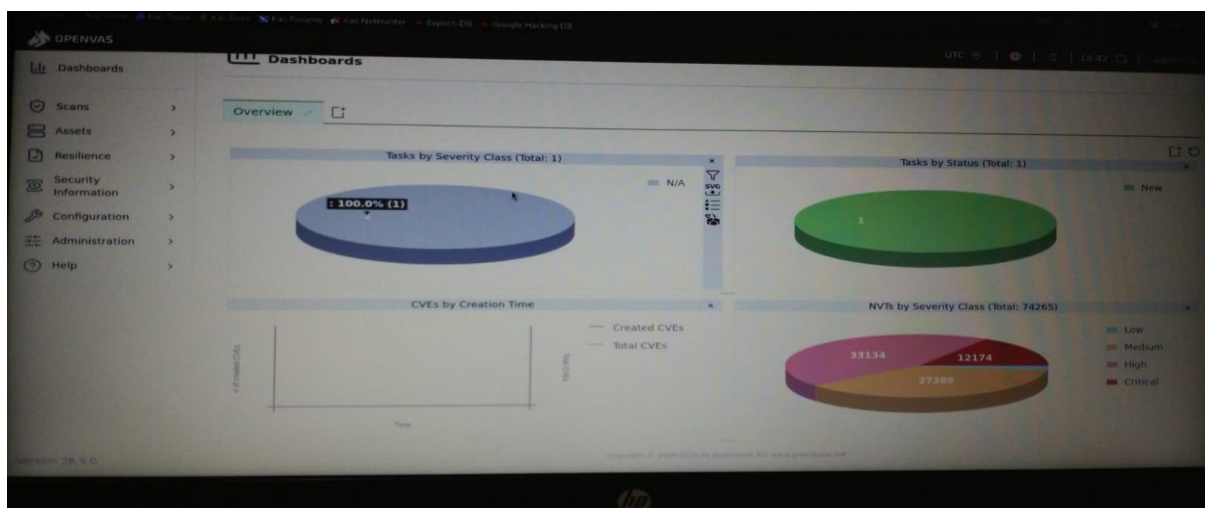
```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.188 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::8565:db8d:31c6:7fe7 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
    RX packets 11 bytes 2288 (2.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 26 bytes 3826 (3.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

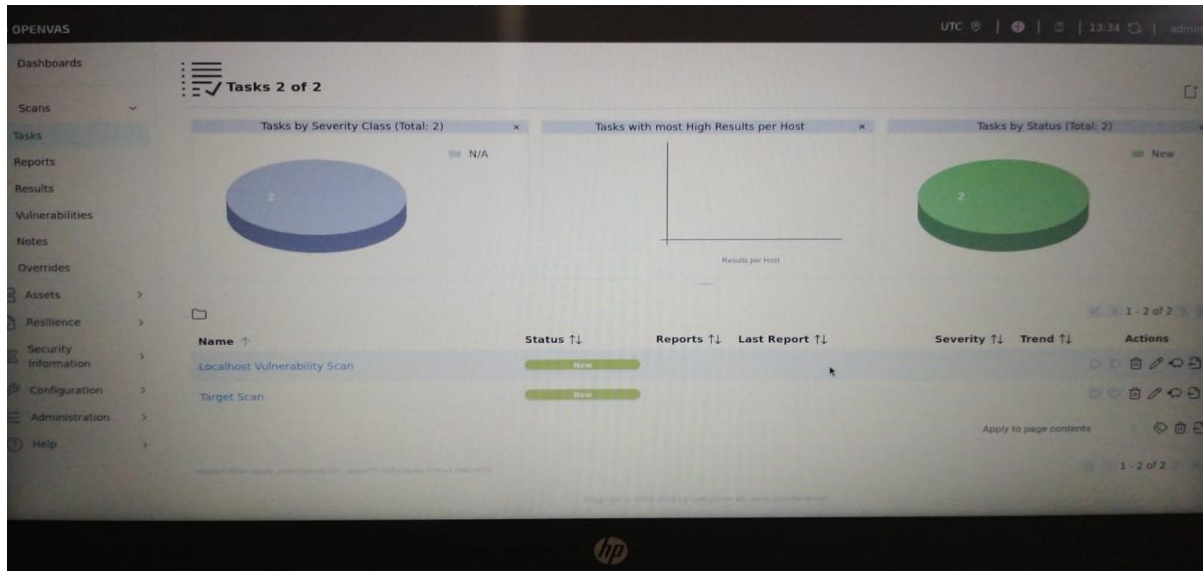
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 37 bytes 2695 (2.6 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 37 bytes 2695 (2.6 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

3.2 OpenVAS Configuration

Although OpenVAS service encountered initialization issues during scanning, dashboard access and target configuration were successfully completed.





3.3 Network Scanning using Nmap

Command Used:

```
nmap -sV 192.168.56.101
```

Findings:

- Port 22 (SSH) – Open
- Port 80 (HTTP) – Open

```
(kali@kali)-[~]
$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 05:59 -0500
Nmap scan report for 192.168.56.101
Host is up (0.000033s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Debian 3 (protocol 2.0)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.13 seconds
```

3.4 Vulnerability Detection using Nmap NSE Script

Command Used:

```
nmap --script vuln 192.168.56.101
```

Observation:

- Service information detected
- No critical CVE found
- System relatively secure



```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ nmap --script vuln 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-26 12:56 -0500
Nmap scan report for 192.168.56.101
Host is up (0.0010s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 16.86 seconds
(kali@kali)-[~]
$
```

3.5 Web Server Scan using Nikto

Tool Used:

- Nikto

Command Used:

```
nikto -h http://192.168.56.101
```

Findings:

- Server information disclosure
- Missing security headers

```
(kali@kali)-[~]
$ nikto -h http://192.168.56.101
- Nikto v2.5.0

+ Target IP: 192.168.56.101
+ Target Hostname: 192.168.56.101
+ Target Port: 80
+ Start Time: 2026-02-26 07:08:14 (GMT-5)

+ Server: Apache/2.4.66 (Debian)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MITM
+ /web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /: Server may leak inodes via ETags, header found with file /, inode: 29cf, size: 63e5b9d733880, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cve
+ OPTIONS: Allowed HTTP Methods: GET, POST, OPTIONS, HEAD
+ /server-status: This reveals Apache information. Comment out appropriate line in the Apache conf file or restrict access to allowed sources. See: 0
+ 8102 requests: 0 error(s) and 5 item(s) reported on remote host
+ End Time: 2026-02-26 07:08:52 (GMT-5) (38 seconds)

+ 1 host(s) tested
```

4. Risk Assessment

Identified Risks

Issue	Likelihood	Impact	Risk Level
SSH Open	Medium	Medium	Medium
Apache Disclosure	High	Low	Medium
Missing Headers	Medium	Low	Low



5. Remediation Recommendations

1. Disable unused services
2. Configure firewall rules
3. Hide Apache version
4. Enable HTTPS
5. Use strong SSH configuration

	A	B	C	D	E	F	G	H	I	J	K	L
1												
2		Issue	Likelihood	Impact	Risk Level							
3		SSH Open	Medium	Medium	Medium							
4		Apache Disclosure	High	Low	Medium							
5		Missing Headers	Medium	Low	Low							

	A	B	C	D	E	F	G	H	I	J	K	L
1	Asset	Vulnerability	CVE	CVSS	Severity	Risk Level						
2	Web Server	Log4j RCE	CVE-2021-44228	10	Critical	High						
3	SMB Server	EternalBlue	CVE-2017-0144	8.1	High	High						
4	HTTPS Server	Heartbleed	CVE-2014-0160	7.5	High	Medium						

6. Conclusion

The vulnerability assessment successfully identified active services and minor configuration risks within the lab environment. No critical vulnerabilities were found due to secure default configuration of Kali Linux.

The practical exercise helped in understanding vulnerability assessment methodology, risk scoring, and professional reporting practices.